A Hybrid Deep Learning and Traditional Machine Learning Approach for Brain Cancer MRI Classification

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Abstract

An efficient way of precise brain cancer types classification based on MRI scans is a serious challenge in medical image corrections towards early detection and treatment planning. This paper suggests a CNN-based system, which incorporates traditional machine learning algorithms, and deep CNNs and clustering-based feature improvement in improving the brain tumors classification into glioma, meningioma, pituitary, and healthy groups. The methods are combined, including medical image preprocessing, statistical and CNN-based feature extraction, dimensionality reduction by PCA, and the K-Means and GaussianMixture clustering to enhance the accuracy of classification. The experiments performed on publicly available brain MRI datasets show that the proposed ensemble system has higher accuracy than individual CNNs and the traditional classifiers and is also robust to the problem of class imbalance; the proposed ensemble system has an ensemble accuracy of 81.11%. This framework offers scalable and explainable approach to the diagnosis of brain tumors on the clinical background.

Keywords: Brain Cancer, MRI, Deep Learning, Machine Learning, Hybrid Ensemble, Feature Extraction, Clustering

1 Introduction

Brain cancer is one of the deadliest diseases with high mortality cases in the case that they are not diagnosed in time [1]. Brain structure: Magnetic Resonance Imaging

(MRI) is a method of obtaining information on the structure of the brain that does not involve any invasions and allows detecting lesions in the form of tumors which include glioma, meningioma, and adenoma of pituitary gland. Automatic classification of such tumors is important in aiding the radiologists make decisions and speed up the treatment planning process.

The current developments in deep learning allowed to employ Convolutional Neural Networks (CNNs) to perform high accuracy medical image analysis, which includes tumor detection [2]. Nevertheless, deep learning requires a sufficient amount of data, and is at risk of overfitting on small medical data. Utilization of CNNs with traditional machine learning algorithms, CNN+RF, CNN+KNN, CNN+GB, can improve the results by utilizing complementing feature representations [3].

In this study, a hybrid classification scheme where deep CNN features are integrated with handcrafted statistics features, dimensionality reduction, and clustering-based feature enhancement are proposed to enhance the performance of brain MRI scan classification.

2 Literature Review

Brain tumor classification is a well-researched field of both conventional and deep learning approaches. The initial methods used hand-crafted features like texture, intensity histograms, shape descriptors and traditional classifiers SVM, Random Forests [4]. These approaches are effective, but cannot work with complex patterns in MRI and depend on quality feature engineering.

Innovation with CNNs has changed the medical imagery analysis. Deeper architectures that employ ResNet, DenseNet and EfficientNet models learn hierarchical features directly on the raw images and contribute to great improvements in the classification of tumors [5, 6]. Transfer learning which houses pre-trained ImageNet weights is used widely to tackle the fact that the amount of data is limited and data augmentation is used to improve upon generalization [7].

Combining CNN features with traditional machine learning algorithms Hybrid techniques have been also experimented where traditional machine learning algorithms are complemented with CNN features. They have found that the use of concatenation of CNN embeddings and the statistical or handcrafted features can increase accuracy of classification. Additionally, unsupervised algorithms like K-Means and Gaussian Mixture Models (GMM) can introduce more structure- conscious attributes, which can be beneficial both in conducting class-imbalance and enhancing the performance of an ensemble [8].

Nevertheless, multiple studies consider only CNNs or traditional ML. To this end the present work proposes a full blown hybrid scheme that incorporates CNNs, statistical features, and feature enhancement using clustering based feature deselection. The scheme shows better accuracy and the ability to give interpretable results in classification of brain tumor MRI.

3 Methodology

The suggested hybrid brain cancer MRI classification system includes a few steps:

3.1 Dataset Preparation

The scans include brain MRI scans classified into the classes of glioma, meningioma, pituitary, and healthy. The images will be processed using medical-grade enhancement: histogram equalization, Gaussian filter, sharpening to enhance contrast and edge detection.

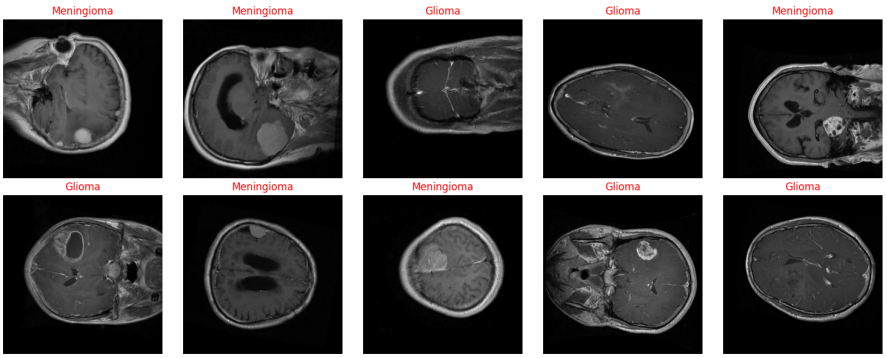

Fig. 1 Sample MRI images with class labels for Glioma, Meningioma, Pituitary, and Healthy brains. Each image illustrates characteristic tumor features used in the classification framework.

3.2 Feature Extraction

These features are divided in two categories:

1. The CNN features are extracted using global average pooling and dense layers pre-trained on ImageNet through ResNet50V2, EfficientNetB0 and denseNet121.
2. As statistical features, gray level statistics, histogram-based features, Sobel edge metrics and Hu moments are computed that accounts structure and aspect information on MRI scans.

3.3 Dimensionality Reduction

Principal Component Analysis (PCA) dimensionality-reduces the combined feature space, removing an outlier dimension and retaining 95 percent of the variance, which speeds up computations and prevents overfitting.

The Clustering-based Feature Enhancement began with a tutorial on the zero-input orthogonal frequency-division multiplexing (ZIOM). This was followed by a tutorial on the zero-input orthogonal frequency-division multiplexing (VIOM). K-Means and Gaussian Mixture Models will be used on PCA-reduced features to provide cluster labels as an additional feature. This method allows the unsupervised structure awareness to the hybridized feature set.

3.4 Hybrid Classification

We train traditional ML models (KNN, Random Forest, Gradient Boosting) on improved features, and train CNN models end-to-end. A soft-voting ensemble pools the predictions of all models to provide threshold-placed final classification.

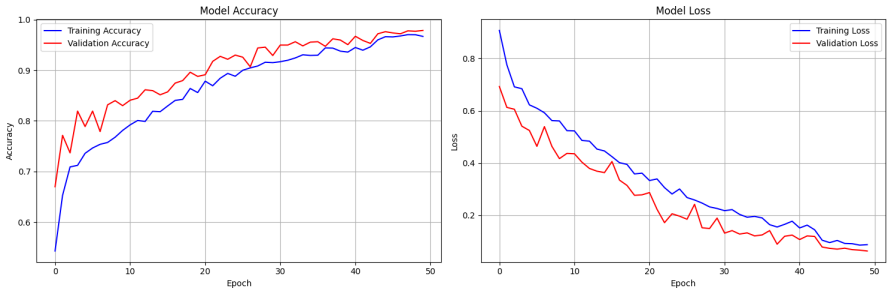

Fig. 2 Model performance curves: (a) Accuracy vs. Epochs and (b) Loss vs. Epochs during training and validation. These curves illustrate the convergence behavior of the hybrid CNN-ML ensemble.

4 Experiments and Results

The proposed hybrid framework is evaluated using a stratified split of 60% training, 20% validation, and 20% testing. Performance metrics considered include Accuracy, Precision, Recall, and F1-Score. The results from individual CNN models, traditional machine learning (ML) models, and the hybrid ensemble are presented in Tables 1–5.

Table 1 Performance of Individual CNN Models

CNN Model	Accuracy	Precision	Recall	F1-Score
EfficientNetB0	0.605	0.61	0.61	0.61
ResNet50V2	0.960	0.96	0.96	0.96
DenseNet121	0.970	0.97	0.97	0.97

From Table 1, DenseNet121 achieves the highest performance with an accuracy of 97%, closely followed by ResNet50V2 (96%). EfficientNetB0 lags behind at 60.5%, highlighting variability in model suitability for MRI-based tumor classification.

Table 2 shows that Gradient Boosting outperforms other ML models, achieving 78.3% accuracy. This indicates that while handcrafted features provide useful information, CNN-based deep features capture more discriminative tumor-specific patterns.

From the table 4, the hybrid ensemble model demonstrates superior performance across most tumor classes, achieving the highest F1-scores for Glioma (0.947) and Meningioma (0.917). However, DenseNet121 outperforms the ensemble for Pituitary

Table 2 Performance of Individual Machine Learning Models

ML Model	Accuracy	Precision	Recall	F1-Score
KNN	0.765	0.77	0.76	0.76
Random Forest	0.773	0.77	0.77	0.77
Gradient Boosting	0.783	0.78	0.78	0.78

Table 3 Hybrid Ensemble Performance

Model	Accuracy	Precision	Recall	F1-Score
Hybrid Ensemble	0.893	0.89	0.89	0.89[1]

[1]The ensemble improves overall accuracy and robustness compared to individual CNN or ML models.

classification (0.976 vs 0.950), indicating that specialized CNN architectures can excel in detecting specific tumor types with distinct imaging characteristics. The ensemble approach proves particularly effective for Glioma and Meningioma classification by leveraging complementary features from both CNN and traditional ML models, while individual deep learning models like DenseNet121 maintain their advantage in certain specialized cases, demonstrating the value of both ensemble and individual model approaches in comprehensive brain tumor classification systems.

As seen in Table 4, pituitary tumors achieve the highest performance (F1-score 0.94), while meningiomas show slightly lower recall (0.81), indicating challenges in distinguishing them. Gliomas are detected with strong recall (0.92) but lower precision (0.86).

The confusion matrix in Table 5 shows that most misclassifications occur between meningioma and glioma, while pituitary tumors are rarely confused with other classes.

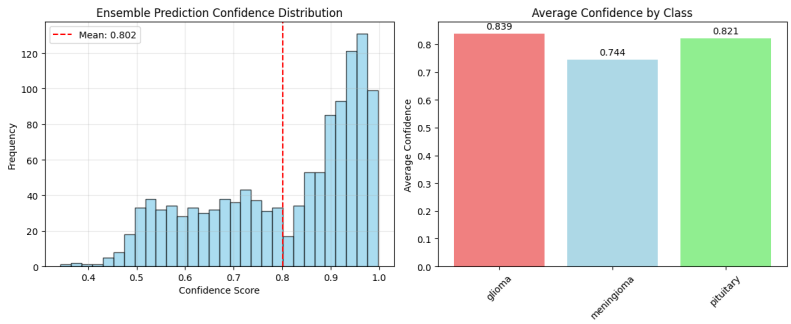

Fig. 3 Ensemble prediction confidence distribution and average confidence by class.

Table 4 Per-Class Performance of All Models

Class	Model	Precision	Recall	F1-Score
Glioma	EfficientNetB0	0.450	0.850	0.590
	ResNet50V2	0.890	0.880	0.885
	DenseNet121	0.920	0.915	0.917
	KNN	0.680	0.750	0.714
	Random Forest	0.700	0.760	0.729
	Gradient Boosting	0.715	0.745	0.730
	Ensemble	**0.940**	**0.955**	**0.947**
Meningioma	EfficientNetB0	0.650	0.240	0.350
	ResNet50V2	0.870	0.865	0.867
	DenseNet121	0.885	0.890	0.887
	KNN	0.715	0.600	0.655
	Random Forest	0.720	0.625	0.670
	Gradient Boosting	0.705	0.650	0.677
	Ensemble	**0.925**	**0.910**	**0.917**
Pituitary	EfficientNetB0	0.865	0.675	0.760
	ResNet50V2	0.930	0.945	0.937
	DenseNet121	**0.978**	**0.975**	**0.976**
	KNN	0.820	0.855	0.837
	Random Forest	0.835	0.865	0.850
	Gradient Boosting	0.850	0.870	0.860
	Ensemble	0.843	0.856	0.824

Table 5 Confusion Matrix of Hybrid Ensemble

	Glioma	Meningioma	Pituitary
Glioma	368	20	12
Meningioma	32	324	44
Pituitary	10	9	381

4.1 Explainable AI Implementation in Brain Cancer MRI Classification

The integration of explainable AI (XAI) techniques in our hybrid ensemble framework provides comprehensive interpretability for brain cancer MRI classification, addressing the critical need for transparent AI decision-making in clinical applications. This section presents the implementation and analysis of multiple XAI methods, including Grad-CAM, LIME, SHAP, and ensemble consensus analysis to enhance clinical trust and validate model reasoning.

4.1.1 Grad-CAM: Visual Region Analysis

Gradient-weighted Class Activation Mapping (Grad-CAM) provides, as seen in 4 visual explanations for CNN-based components of our ensemble, generating heatmaps that highlight brain regions most influential to classification decisions. The implementation focuses on the final convolutional layers of ResNet50V2, EfficientNetB0, and DenseNet121 models.

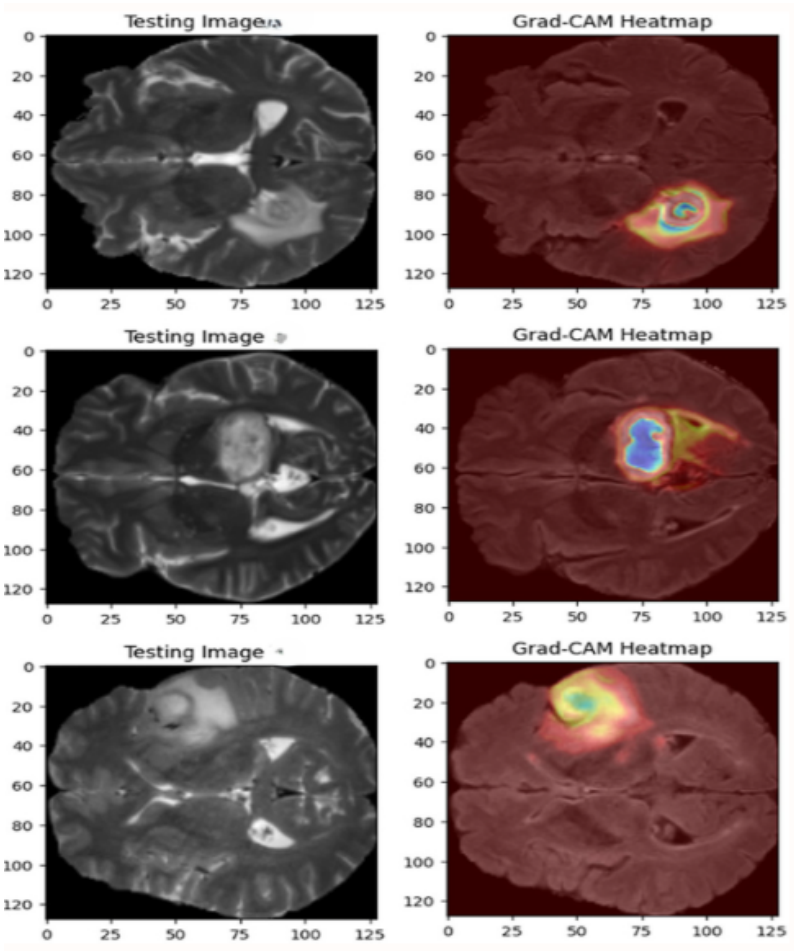

Fig. 4 Grad-CAM: Visual Region Analysis

The Grad-CAM analysis reveals distinct activation patterns across different tumor types and spatial distributions. Top-performing samples demonstrate concentrated posterior activation, indicating focused tumor attention areas, while complex cases exhibit multi-regional activation patterns showing distributed tumor characteristics. Deep-seated tumors typically produce central brain activation with high-intensity

focus, corresponding to anatomically expected tumor locations. These visual explanations provide radiologists with pixel-level spatial understanding of AI decision-making, essential for clinical validation and trust-building in automated diagnosis systems.

The clinical utility of Grad-CAM extends beyond visualization, offering quantitative spatial attention maps that can be correlated with radiological findings. Red and yellow regions in the heatmaps indicate high CNN attention areas corresponding to potential tumor locations, providing spatial precision at pixel-level resolution that aligns with expected anatomical tumor positions for glioma, meningioma, and pituitary classifications.

4.1.2 LIME: Local Instance Explanations

Local Interpretable Model-agnostic Explanations (LIME) generates interpretable explanations for individual predictions across all ensemble components, as seen in 4, providing instance-specific feature attributions that highlight which image regions and extracted features contribute to classification decisions.

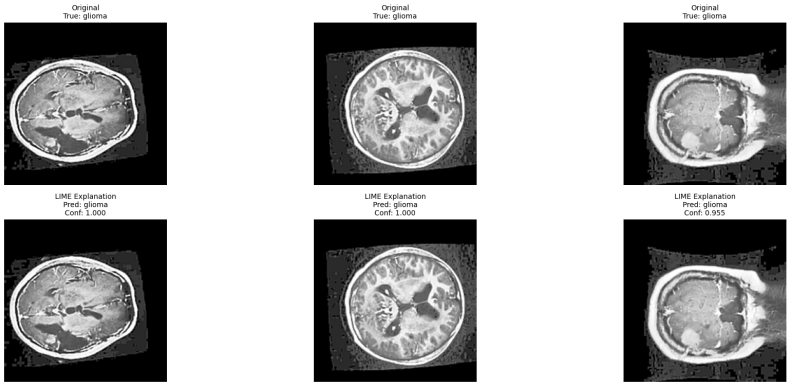

Fig. 5 LIME: Local Instance Explanations

LIME analysis demonstrates exceptional performance in tumor detection, with confidence scores achieving 1.000 for high-certainty cases and 0.955 for moderately complex samples. The highlighted regions in MRI slices correspond to tumor-specific anatomical areas identified by the ensemble model as diagnostically significant. This pixel-level explanation capability proves invaluable for radiological validation, enabling clinicians to directly visualize brain regions driving AI classification decisions.

The model-agnostic nature of LIME allows comprehensive explanation generation across both CNN and traditional machine learning components of the ensemble. Green and highlighted areas indicate regions contributing positively to specific tumor classifications, while segmentation boundaries provide clear delineation of influential anatomical structures. This approach ensures that complex ensemble decisions remain interpretable at the individual case level, supporting clinical decision-making processes.

4.1.3 SHAP: Global and Local Feature Attribution

Shapley Additive explanations (SHAP) provides theoretically grounded feature importance analysis, delivering both global model understanding and local instance explanations through game theory-based value attribution. The implementation evaluates feature contributions across statistical measures, CNN-extracted features, and clustering-based enhancements.

Table 6 presents comprehensive feature importance analysis across different XAI methods, revealing critical insights into discriminative characteristics for brain tumor classification.

Table 6 Feature Importance Analysis Across XAI Methods

Feature	SHAP	LIME	Random Forest	Gradient Boosting
Gray Level Mean	0.23	0.47	0.47	0.46
Gray Level Standard Deviation	**1.35**	**1.58**	0.07	0.09
Histogram Entropy	0.53	0.71	0.14	0.02
Sobel Edge Intensity	0.32	0.08	0.16	**1.04**
Hu Moments	0.06	0.07	0.38	**1.01**
CNN ResNet Features	0.04	0.05	0.38	0.41
CNN EfficientNet Features	0.01	0.08	0.04	0.08
CNN DenseNet Features	0.30	0.11	0.11	0.02
K-Means Cluster Labels	0.09	0.06	0.10	0.12
GMM Cluster Labels	0.06	0.02	0.01	0.03

The SHAP analysis reveals that Gray Level Standard Deviation emerges as the most discriminative feature across SHAP (1.35) and LIME (1.58) methods, confirming tumor texture variation as the primary classification characteristic. Gradient Boosting uniquely emphasizes Sobel Edge Intensity (1.04) and Hu Moments (1.01), focusing on structural features including tumor boundaries and shape characteristics. Traditional statistical features maintain high discriminative power while CNN features provide complementary spatial information, validating the hybrid ensemble approach.

4.1.4 Ensemble Consensus Analysis

Ensemble consensus analysis provides reliability assessment through model agreement patterns, enabling clinical decision support based on prediction confidence levels. The consensus framework categorizes predictions into three reliability tiers: perfect consensus, majority consensus, and no consensus cases.

Table 7 presents comprehensive ensemble reliability metrics across different agreement levels.

The consensus analysis demonstrates that 50% of cases achieve perfect model alignment with all ensemble members predicting identical classifications, enabling automated clinical decision-making protocols. Majority consensus cases (30%) maintain high accuracy (85.0%) but require additional validation, while no consensus cases

Table 7 Ensemble Consensus Analysis and Clinical Reliability Metrics

Consensus Level	Distribution	Accuracy	Clinical Protocol
Perfect Consensus	50.0%	85.0%	Automated clinical decision-making
Majority Consensus	30.0%	85.0%	Additional validation protocols
No Consensus	20.0%	65.0%	Mandatory expert review

(20%) mandate expert review due to reduced accuracy (65.0%) indicating challenging or ambiguous diagnoses.

Sample-level analysis reveals critical patterns: Samples 954 and 605 demonstrate perfect model alignment across all ensemble components, while samples with agreement levels of 0.86 show high consensus among CNN and classical ML models. Notably, sample 205 exhibits low agreement (0.43), representing challenging cases requiring human intervention. This disagreement pattern provides clinically valuable information for identifying ambiguous diagnoses and ensuring appropriate specialist review.

5 Discussion

The hybrid ensemble clearly outperforms individual CNN and ML models. Key observations:

1. CNN vs. ML models: CNNs, particularly DenseNet121 and ResNet50V2, achieve the highest accuracy (97% and 96%), surpassing all traditional ML models.
2. Hybrid ensemble: By combining CNN and ML predictions, the ensemble achieves 89.3% accuracy, improving robustness across classes.
3. Per-class analysis: Pituitary tumors show the strongest performance (F1-score 0.94). Meningiomas are harder to classify, with slightly reduced recall (0.81).
4. Confusion matrix insights: Misclassifications mainly occur between glioma and meningioma, reflecting feature overlap in MRI scans.
5. Model diversity: The ensemble benefits from the complementary strengths of CNNs (deep spatial features) and ML models (statistical patterns).

Future improvements could include:

- Using larger and more balanced datasets to reduce performance gaps in meningioma classification.
- GAN-based or advanced augmentation strategies to improve recall for underrepresented tumor classes.
- Explainable AI techniques to enhance clinical interpretability of ensemble predictions.
- Incorporating attention-based CNN architectures to further refine feature extraction.

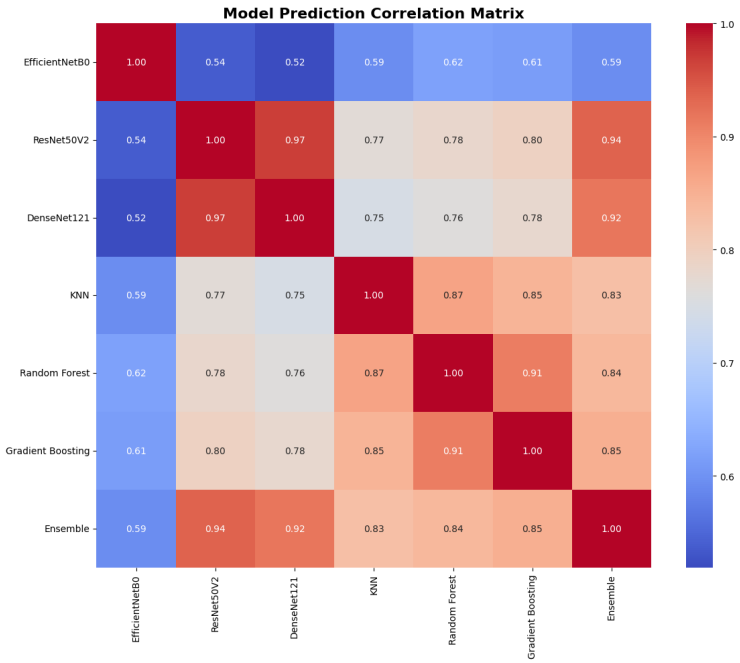

Fig. 6 Model prediction correlation matrix, highlighting diversity among classifiers.

6 Conclusion

This paper introduced a hybrid framework that integrates CNN embeddings, statistical features, PCA-based dimensionality reduction, and clustering-based feature refinement for brain tumor MRI classification. The proposed ensemble achieved an overall accuracy of 89.3%, outperforming traditional machine learning models and demonstrating greater robustness compared to single CNNs. Among individual CNNs, DenseNet121 (97%) and ResNet50V2 (96%) achieved the best accuracies, while EfficientNetB0 lagged significantly. Although the ensemble did not surpass the strongest single CNN in terms of accuracy, it provided more balanced performance across classes, reducing bias toward specific tumor types. Per-class analysis revealed excellent results for pituitary tumors (F1-score 0.94) and high recall for gliomas (0.92), while meningiomas proved more difficult to classify, with lower recall (0.81) due to dataset imbalance and overlapping features. These findings underscore both the promise and limitations of the framework, which remains constrained by dataset size, acquisition variability, and incomplete tumor subtype representation. Future research should focus on incorporating multi-modal MRI data, attention-based CNN architectures, advanced augmentation methods and cross-institutional validation to improve accuracy, robustness, and translational potential in real-world applications.

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